

but it would ultimately be affected by the layer being etched.

Q. Could you tell us about some benefits that are enjoyed by today's design tool users?

A. Companies have discovered in the past year that, by using Tencent yield enhancement, they can save tens of millions of dollars in cost of components, by rapidly identifying defects in the layout that affect their yield. They are able to analyse every chip that fails in test and compare it to the layout. This lets them figure out if it is a systematic or random issue, following which maths tools analyse and bring out correlation.

The next is that, without emulation, most networking companies would have been unable to continue as complexity of data networking is now so great that simulation is no longer possible.

Q. Are there any similar examples from the foundry side?

A. The foundry industry has found that, by requiring their customers to use a tool called Calibre PERC, they could assure that customers' designs had protection for electrostatic discharge (ESD).

What previously happened was that every customer had his or her own different techniques to check. As a result, when wafers came back, ESD destroyed those wafers and these customers would tell foundries that the yield was low.

Q. What are the new tools coming up in the automotive sector?

A. New tools are now available that enable embedded software development capabilities for automotive design using standards such as automotive open system architecture (AUTOSAR). We are now also seeing new analysis tools for connectivity in a car, as well as tools for documentation and even to help technicians with wiring.

For electronics manufacturing ser-

vices, design for manufacturing and assembly markets we see that, as design complexity increases, these companies start facing issues in library management. So that is another area where problems can be solved.

**ENGINEERS
WORKING ON BIG
DIGITAL CHIPS NOW
HAVE TO ADOPT
EMULATION TO
JUST HANDLE THE
ENORMOUS AMOUNT
OF VERIFICATION
NEEDED**

Q. How has an increase in digital chip complexity affected the design process?

A. Engineers working on big digital chips now have to adopt emulation to just handle the enormous amount of verification needed. To help in solving these new challenges, we saw the introduction of integrated simulation and emulation as well as a big advance in power analysis introduced in cooperation with ANSYS.

Additionally, it is now also possible to design for test in the emulator itself.

Q. What is the roadmap for system design tools, and how does it compare to chip design tools?

A. Traditional chip design tools are growing at the same rate as the semiconductor industry. However, since system design is at an early stage, it is growing at two to three times as much.

Looking at the systems-side roadmap, we can expect design tools made to handle the integration of optics and computer systems, backplanes and printed circuit boards. **EFY**

**Share Your
Knowledge**

If you are an expert in your field of electronics, why not share your knowledge with the world?



Electronics For You invites electronics professionals and industry experts to write articles on their area of expertise and interest from different perspectives

Some of the topics we want you to write about are:

- Photonics: Optics & Optoelectronics
- Automation
- Power Electronics
- Components
- Telecom

So if you have an interesting topic or original article to share, please let us know at efyeditorial@efy.in

We would love to hear your good ideas too!

electronics
FOR YOU